

Claire Kim

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Education

University of California, Los Angeles (UCLA) | Bachelor of Science in Mechanical Engineering **Sep 2025 – Jun 2029**

Projects

Product Engineer

Jul 2026 – Sep 2026

Matcha-matic | Independent Project

Camarillo, CA

- **Mechanical Design & Prototyping:** Engineered a compact, low-cost (\$50 budget) automated matcha maker from scratch in SolidWorks, utilizing FDM 3D printing and iterative design for portable on-the-go functionality
- **Mechanism & Fluid Integration:** Engineered a magnetic stirrer powertrain using a brushed DC motor, dual side magnets, and custom aerating attachments to simultaneously drive fluid mixing and micro-foam generation
- **Mechatronics & Embedded Control:** Integrated an ESP32 microcontroller, motor driver, and silicone-tube water distribution system to execute multi-mode automated programs via a 4-button user interface

Mechanical Engineering Lead

Apr 2026 – Jun 2026

Petalux | Creative Labs

Los Angeles, CA

- **Mechanical Design & Prototyping:** Engineered the mechanical structure of an automated kinetic lamp in Autodesk Fusion 360 from requirements to fully constrained CAD assemblies and prototypes
- **Mechanism & Actuation:** Designed a custom powertrain with central screw linkage and specialized petal gears to actuate a 10-petal opening sequence while minimizing axial wobble through tight tolerances
- **Cross-Functional Integration:** Developed space-efficient internal chassis routing for electronics and wiring without obstructing moving parts, while partnering with cross-functional teams to integrate user-interface hardware via snap-fits and custom cutouts

Mechanical Engineering Lead

Oct 2025 – Nov 2025

Aeris | Theta Tau

Los Angeles, CA

- **System Architecture:** Directed “Aeris”, a smart-cooling system integrating facial recognition and tri-axis actuation to achieve dynamic, automated airflow control
- **Enclosure Design & DFA:** Engineered linkages and custom enclosures in SolidWorks and Fusion 360 using Design for Assembly (DFA) principles, streamlining Arduino hardware integration via high-precision friction fits
- **Prototyping & Validation:** Executed the full product lifecycle from FDM 3D printing to electromechanical validation, performing precision soldering and iterative troubleshooting to eliminate hardware errors

Chassis Mechanical Engineer

Sep 2025 – Jun 2026

BB-8 | X1 Robotics - American Society of Mechanical Engineers

Los Angeles, CA

- **Mechanism Design:** Designed and assembled BB-8 drive mechanisms in SolidWorks, optimizing weight distribution and omnidirectional motion stability
- **Electromechanical Packaging:** Positioned motor housings and drivetrain components to fit within a strict 2mm machined shell tolerance while maximizing internal volume
- **Structural Analysis:** Modeled the core support ring in CAD, validating structural stress tolerances and in-house manufacturing feasibility for rapid fabrication

Experience

Micro and Nano Manufacturing Lab

Sep 2026 – Present

Undergraduate Researcher

Los Angeles, CA

- **Hardware Redesign:** Redesigning the portable EWOD (electrowetting-on-dielectric) chip and housing device, implementing targeted physical modifications to transition the device toward consumer market readiness
- **Usability Optimization:** Restructuring physical hardware components to improve user ergonomics, streamlining the portable system layout to ensure the platform is accessible and intuitive to operate

Technical Skills / Certifications

- **Modeling:** SolidWorks, Fusion360, FEA, AutoCAD, DFM, Tolerance Analysis, Rapid Prototyping
- **Electronics & Manufacturing:** Arduino, Soldering, Machining, Welding, 3D Printing, CNC
- **Programming Languages:** Python, C++, JavaScript, CSS/HTML
- **Relevant Coursework:** Statics, Engineering Design/CAD, MATLAB Programming
- **Certifications:** CWSA (Solidworks), OSHA (Safety)